Three Stage Dynamic Context Engineering for LLM Tool Usage

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Background

- Top models aren't perfect at calling tools
- Most approaches increase performance with model changes or methods like reasoning / prompting
- However, methods like **Reflexion** and **ACE** show promise for in-context learning of tool use by letting LLMs leave memory notes for future improvements

Rank	Overall Acc	Model
1	72.01	GLM-4.5 (FC)
2	71.21	Claude-Opus-4-1-20250805 (FC)
3	68.91	GLM-4.5-Air (FC)

Top three models on the Berkeley Function Calling Benchmark, all under 75% accuracy

	Reflexion	ACE	DCE (us)
Uses only relevant, tool-specific memory	❌	❌	✅
Unlimited memory store	❌	❌	✅
Can train with ground truth feedback	🟨 (only binary)	✅	✅

Problem

Existing in-context techniques use a monolithic memory store of path learnings, causing…

- **Contextual Clutter**: Irrelevant tool guidance interferes with task.
- **Uncontrolled Growth**: Memory becomes too large (costly, slow to process, may exceed context window)

Methods

We introduce a new approach to in-context tool learning—Dynamic Context Engineering (DCE). It consists of two stages: **Learning** and **Evaluation**

DCE Learning Process - creates playbook by reflecting on errors

DeepSeek-V3.2-Exp

Generator (LLM output)

BFCL Evaluation Framework

Ground Truth Evaluation

DeepSeek-V3.2-Exp or Claude-4.5-Haiku

Reflector (how to improve?)

error_identification, root_cause_analysis, correct_approach, key_insight, reasoning

DeepSeek-V3.2-Exp or Claude-4.5-Haiku

Curator (how to edit memory?)

Memory Section: event_settings_api

Operation: *ADD*, MODIFY, DELETE

Delta: Use restore_mobile_app_alert_config for restoration operations, not find functions.

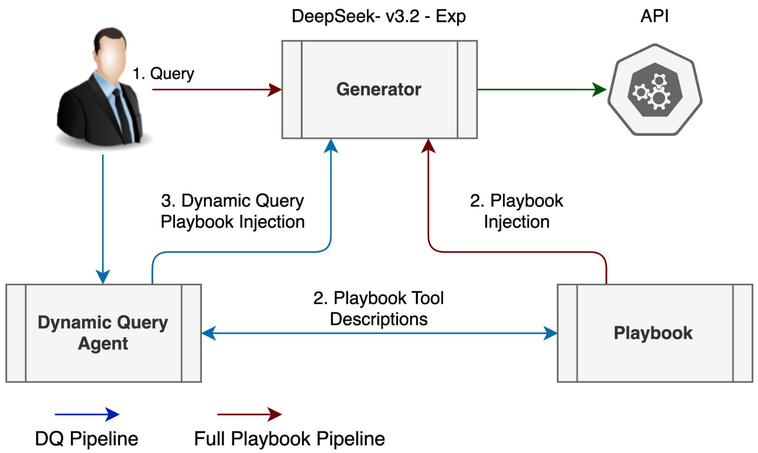
Playbook (memory)

Section 1: event_settings_api

→ **Delta1**: Use get_event_specification_infos_by_ids for unified event specification lookup, not separate built-in/custom functions

→ **Delta2**: Use restore_mobile_app_alert_config for restoration operations, not find functions.

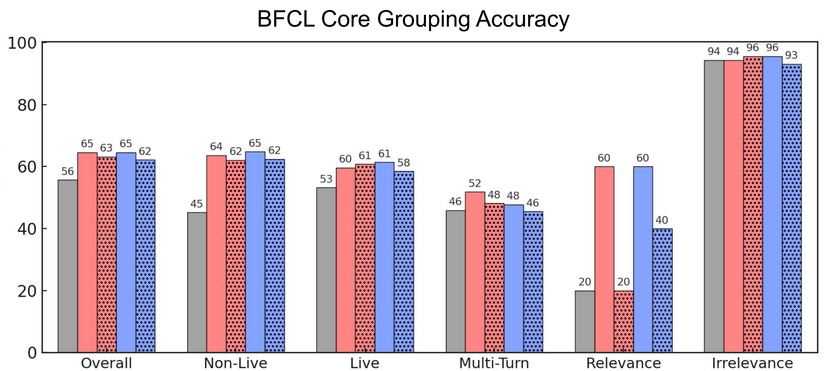
DCE Evaluation Process - using playbook on new tool tasks

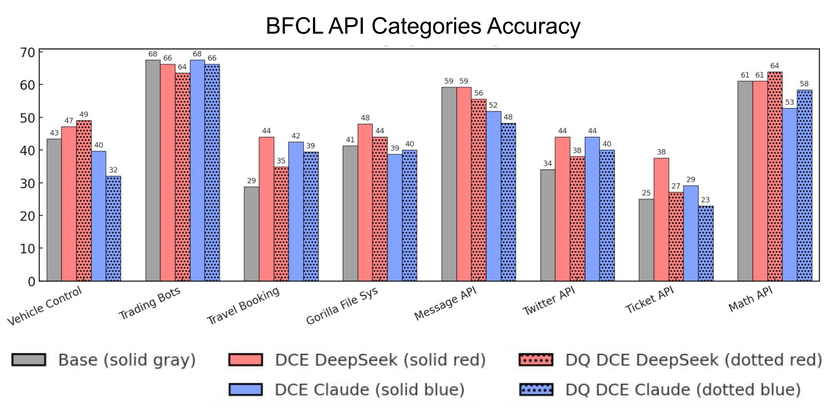


Runs in EITHER "Playbook Injection" mode or "Dynamic Query Playbook Injection" mode

Experiments

- We run baseline and DCE on BFCLv3's tool calling benchmark
- To create playbook, we train on a sample of 1866 from BFCL
- We test on 934 samples from BFCL, with and without Dynamic Query (DQ)





Analysis

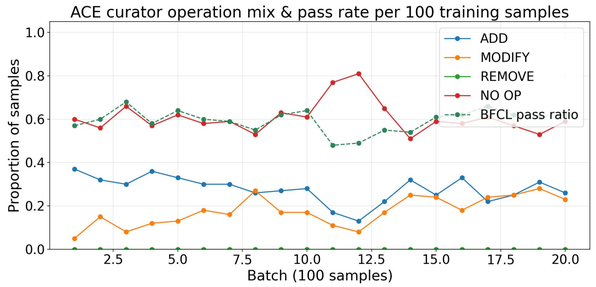
- Despite being weaker on its own, **Haiku builds a playbook that gives almost the same accuracy** as the DeepSeek-based playbook.

Playbook (memory)

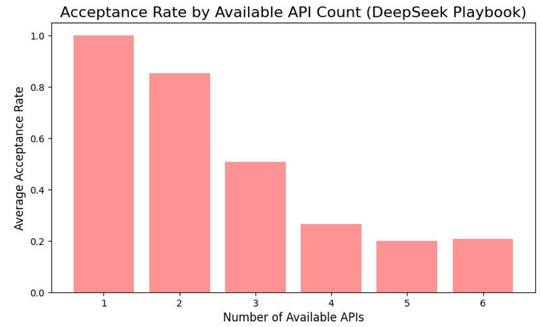
Section: get_sport_team_details

DeepSeek Delta1: Detect irrelevance - avoid calls for queries outside team details scope like player stats

Claude Delta1: Scope: team roster, stats, history only. Decline player-specific queries (e.g., top scorer, individual stats).



- Frequency of **MODIFY starts low** and **increases as playbook matures**
- Lack of REMOVE operations



- Dynamic query: **more selective** as **number of available tools increases**
- Small drops in accuracy

Conclusion

Structured, tool-specific memory works: Our DCE playbook turns past failures into editable, per-API rules (ADD/MODIFY/DELETE), giving the model precise, non-interfering guidance for each tool.

Reliable learning without huge models: One pass of DCE training yields ~10% better tool-calling accuracy, and even smaller reflection models can curate playbooks that strong generators use effectively.

Scales to larger systems: Dynamic Query shows we can retrieve only relevant playbook sections and keep most of the benefit, pointing toward scalable, memory-aware tool use in real deployments.

References

Patil, S. G., Mao, H., Ji, C. C., Yan, F., Suresh, V., Stoica, I., & Gonzalez, J. E. (2025). *The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models.* Proceedings of the 42nd International Conference on Machine Learning (ICML).

Zhang, Y., Chevalier, A., Ajith, A., & Chen, D. (2025). *Agentic Context Engineering (ACE): Evolving Playbooks for Self-Improving LLMs.* arXiv preprint arXiv:2510.04618.